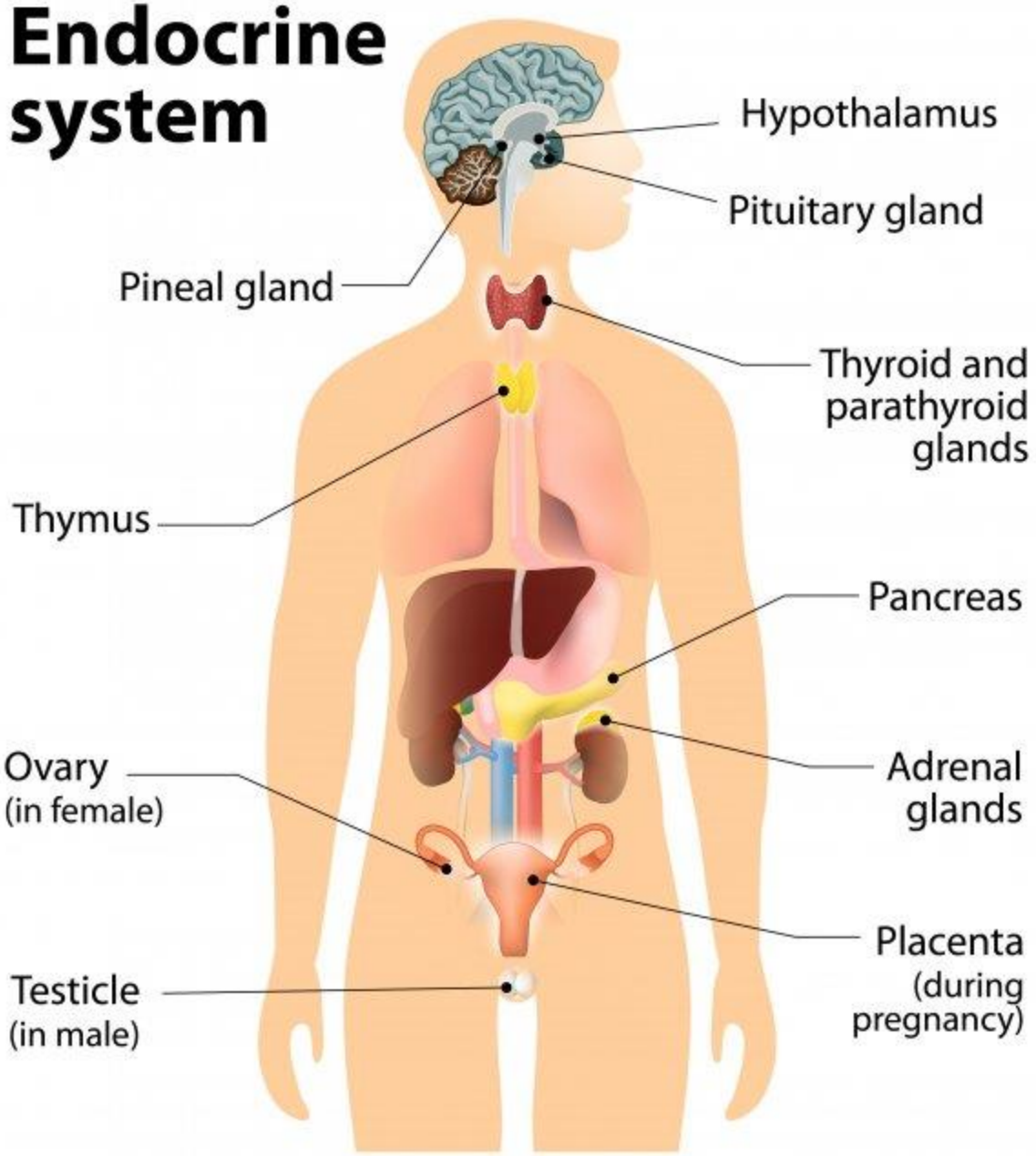
Endocrinology and development

Endocrine system



Pituitary Gland

- The pituitary gland, also known as the hypophysis.
- It is located at the base of the brain, attached to the hypothalamus, superiorly, by a stalk called the infundibulum.
- The pituitary is about 1 cm in diameter.
- The pituitary gland's appearance is that of a pea on a stalk.
- It is divided into anterior pituitary and posterior pituitary lobes.

Anterior pituitary gland

- The anterior pituitary gland is also called the adenohypophysis and is composed of glandular tissue.
- The anterior lobe has three regions:

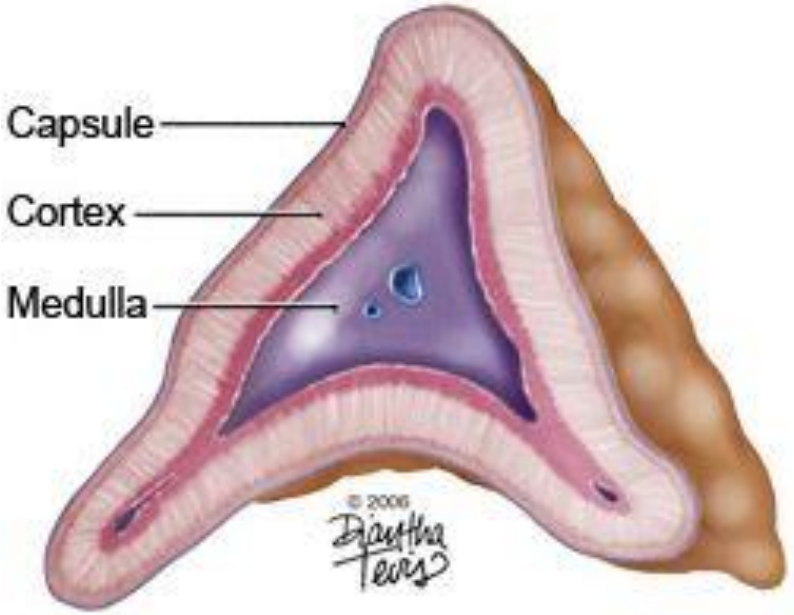
1. Pars distalis: The largest, most anterior part of the pituitary gland.
2. Pars tuberalis: The extension that wraps around the adjacent area of the infundibulum.
3. Pars intermedia: The slender and narrow band that borders the posterior lobe of the pituitary gland.

ADRENAL CORTEX

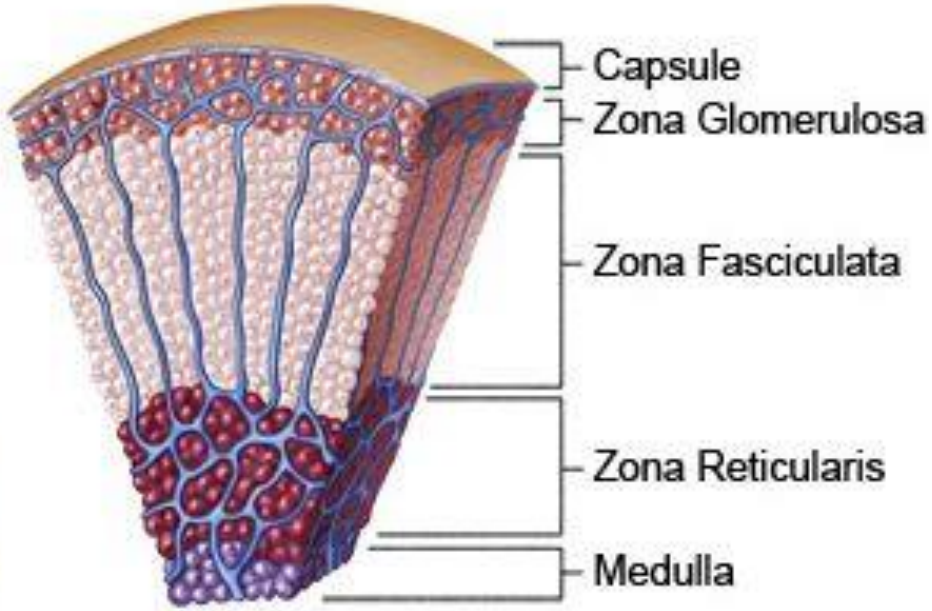
- The adrenal glands are two small organs, located above the kidneys.
- Each adrenal consists of two distinct tissues
  - an outer cortex (with 3 zones)
  - inner medulla

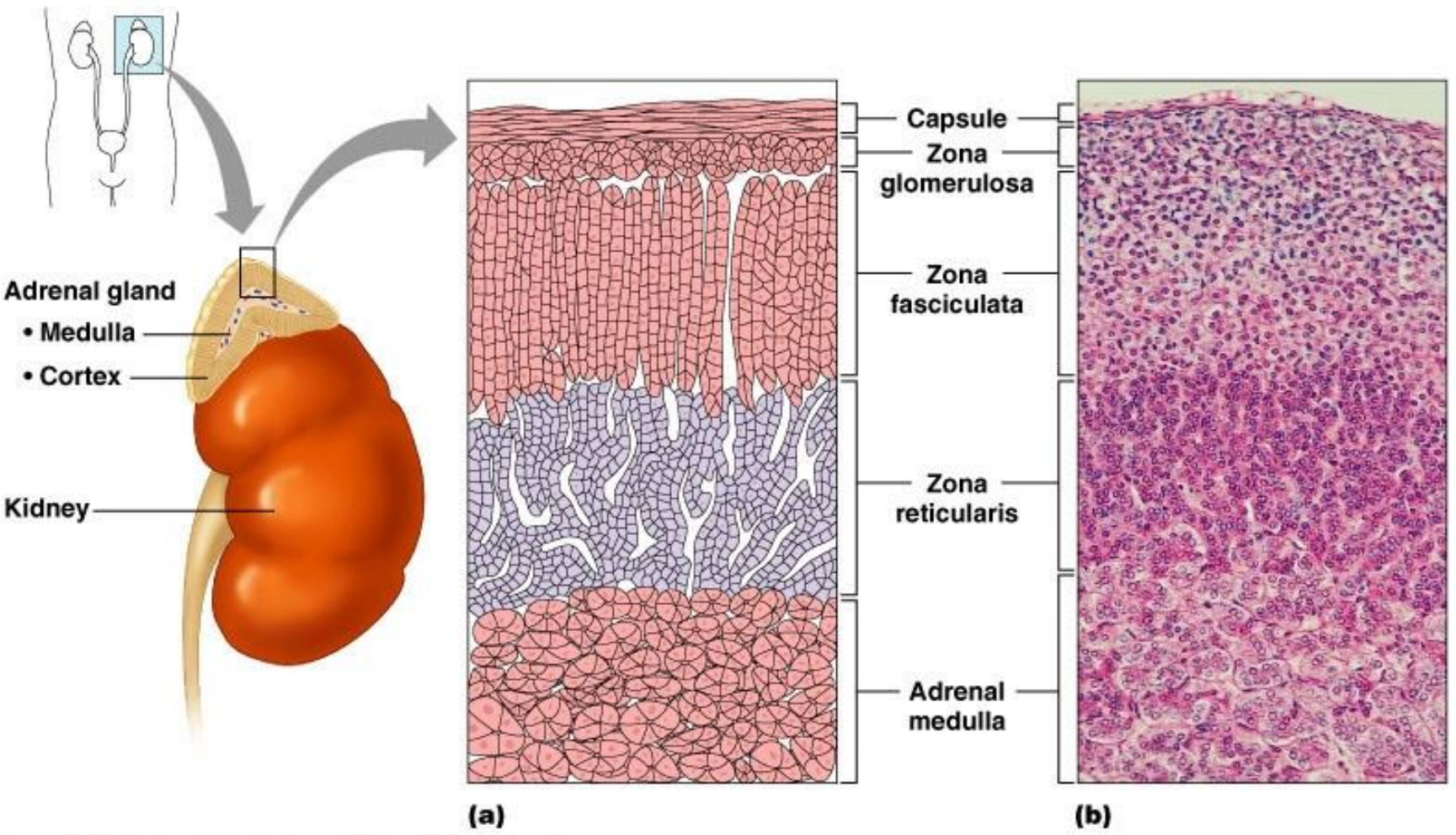
Adrenal Gland Cross Sections

Transverse Section



Microscopic Section





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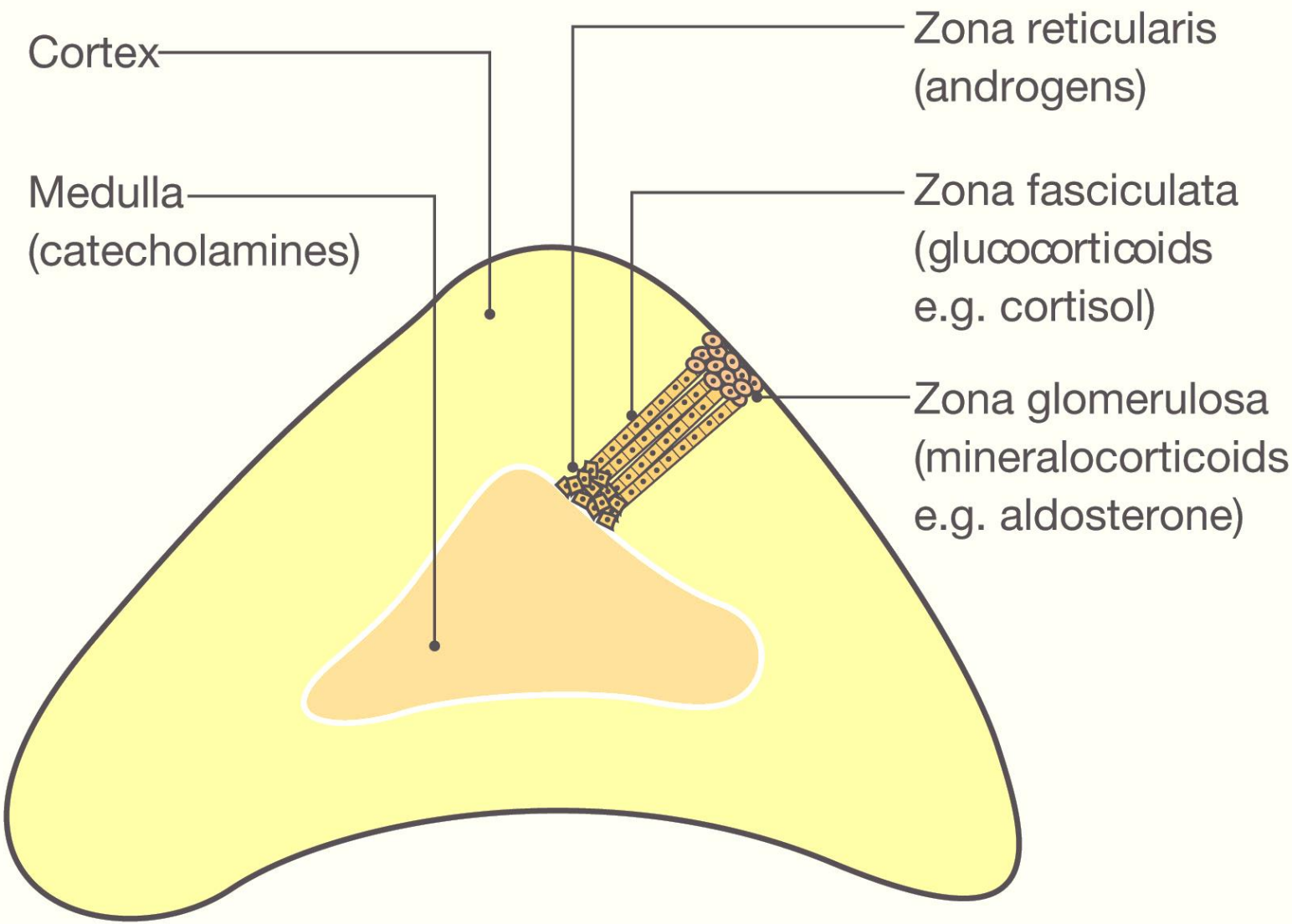
Adrenal hormones

- Around 50 steroid hormones as adrenocorticosteroids are produced by adrenal cortex.
- But, only a few of them possess biological activity.
- Adrenocorticosteroids are classified into three groups according to their biological action.
  - Glucocorticoids
  - Mineralocorticoids
  - Androgens and estrogens
- Inner medulla produce catecholamines epinephrine and norepinephrine.

Androgens and estrogens

- Androgens are 19-carbon and estrogens are 18-carbon containing steroids.
- The innermost adrenal cortex zona reticularis produces small quantities of androgens and estrogens.
- These hormones affect sexual development and functions.
- They are mostly produced by gonads.
- Dehydroepiandrosterone is a precursor for androgens which is synthesized in adrenal cortex.

Structure of the adrenal gland



Hormones produced by different layers given in parentheses.

Gland & region/ cells	Hormones	Regulation of secretion	Functions
Adrenal cortex Zona glomerulosa	Mineralcorticoids, e.g. aldosterone	Stimulated by angiotensin II	Regulates salt & water balance in blood by increasing Na^+ & H_2O absorption and K^+ secretion by the distal convoluted tubules in the kidney
Adrenal cortex Zona fasciculata	Glucocorticoids, e.g. cortisol & weak androgens	Stimulated by adrenal corticotrophic hormone	Suppresses immune response and regulates carbohydrate metabolism
Adrenal cortex Zona reticularis	Weak androgens, e.g. dehydroepiandrosterone	Stimulated by adrenal corticotrophic hormone	Precursor for testosterone production
Adrenal medulla Chromaffin cells	Catecholamines, e.g. Epinephrine & norepinephrine	Preganglionic sympathetic neurons	Increases heart rate, respiration, and blood pressure Constricts vessels to reduce blood flow to GI tract

Adrenal medulla

- Adrenal medulla produces two important hormones
  - epinephrine (formerly adrenaline)
  - norepinephrine (formerly noradrenaline)
- Epinephrine is a methyl derivative of norepinephrine.
- Dopamine is another catecholamine, produced as an intermediate during the synthesis of epinephrine.
- Norepinephrine and dopamine are important neurotransmitters in the brain and autonomic nervous system.
- Catecholamines are produced in response to fight, fright and flight.
- These include the emergencies like shock, cold, fatigue, emotional conditions like anger etc.

HORMONES OF GONADS

- The gonads (testes in males, ovaries in females) synthesize sex hormones.
- The steroid sex hormones are responsible for growth, development, maintenance and regulation of reproductive system.
- The sex hormones are categorized into three groups:
  - Androgens
  - Estrogens
  - Progesterone